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Data driven Investment strategies

אסטרטגיות השקעה מבוססות נתונים

פרויקט מחקרי

דוח סקירה ספרותית

הוכן לשם השלמת הדרישות לקבלת

תואר ראשון במדעי הטבע B. Sc.

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הוגש למחלקה למדעי המחשב

המכללה האקדמית להנדסה סמי שמעון

באר-שבע

תאריך עברי   תאריך לועזי

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חתימת סטודנט 1:__________________________ תאריך: ___________________________

חתימת סטודנט 2:__________________________ תאריך: ___________________________

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אישור ועדת הפרויקטים:____________________ תאריך: ________________________

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Abstract:

This project aims to explore and deeply understand widely used risk-adjusted performance measures, including the Sharpe, Sortino, Treynor, Calmar, and Information ratios. Rather than evaluating these metrics solely through numerical computation, the project focuses on analyzing the theoretical assumptions behind each ratio, understanding their strengths and limitations, and identifying conditions in which they may fail to properly characterize financial risk. As the project is currently in its early stages, the present work outlines our research objectives and methodological direction rather than reporting empirical results.

In addition to studying existing ratios, we plan to investigate the potential for constructing a new risk-adjusted measure based on the stability of compounding returns. Because financial prices evolve through multiplicative processes, return sequences often approximate exponential growth. By fitting an exponential regression model to log-transformed price data, we intend to measure how closely an asset's pricing behavior aligns with a smooth exponential compounding structure. This approach differs from trend- or momentum-based indicators: instead of identifying direction, it evaluates the consistency of return generation over time. Future work will empirically assess how this proposed measure compares to classical ratios.

1. Introduction:

Risk-adjusted performance measures play a critical role in financial decision-making. They allow investors to compare assets whose raw returns may differ substantially by incorporating measures of risk-whether total volatility, downside volatility, systematic risk, or drawdown behavior. These ratios lie at the heart of portfolio construction, risk management, and performance evaluation of both individual assets and investment strategies.

However, despite their widespread use, traditional risk-adjusted ratios rely on simplified assumptions that do not always reflect real-world market dynamics. Volatility is often treated as a symmetric measure of risk, even though negative and positive deviations do not carry equal meaning to investors. Market regimes shift, distributions become fat-tailed, and compounding behaviors introduce structural patterns that are not explicitly addressed by classical ratios. As a result, these measures may produce misleading signals under certain conditions.

The primary goal of this project is to develop a deep understanding of these ratios-mathematical, conceptual, and practical. We aim to examine how each metric behaves, the assumptions embedded in its structure, and the types of market environments in which it succeeds or fails. Beyond this conceptual analysis, we plan to introduce a new perspective on risk-adjusted measurement by examining the stability of exponential compounding, a natural property of financial return series.

At this early stage, our work focuses on shaping the research questions, establishing methodological foundations, and outlining the theoretical motivation for the proposed exponential-fit measure.

2. Literature Review

2.1. Background:

Risk-adjusted performance measures are designed to evaluate investment outcomes by balancing return and risk. Over time, a wide range of such ratios has been proposed in the financial literature, each addressing a different aspect of risk or performance evaluation. The existence of multiple risk-adjusted ratios reflects the complexity of financial risk rather than a deficiency of any single metric.

- **Sharpe Ratio (Sharpe, 1966):** The Sharpe ratio measures how much excess return an asset generates for each unit of total risk (volatility) it takes. It evaluates whether the return justifies the variability in the asset's performance.

$$Sharpe = \frac{R_p - R_f}{\sigma_p}$$

- **Sortino Ratio (Sortino & Price, 1994):** The Sortino ratio refines the Sharpe ratio by penalizing only downside volatility the kind of volatility investors actually care about.

$$Sortino = \frac{R_p - R_f}{\sigma_d}$$

- **Treynor Ratio (Treynor, 1965):** The Treynor ratio measures how much excess return an investment generates for each unit of systematic risk the risk that comes from the overall market.

$$Trenor = \frac{R_p - R_f}{\beta_p}$$

- **Information Ratio (Grinold, 1989):** The Information ratio measures how consistently an investment beats a benchmark, adjusted for the volatility of that difference.

$$Informatio = \frac{R_p - R_b}{\sigma_{(R_p - R_b)}}$$

- **Calmar Ratio (Young, 1991):** The Calmar ratio compares annual return to the maximum drawdown - the worst peak-to-trough loss.

$$Calmar = \frac{R_p - R_f}{D_{max}}$$

Each ratio is built on specific assumptions and is suited to market conditions. As a result, every measure exhibits both strengths and limitations, depending on how risk is defined and which characteristics of asset behavior are emphasized. Different ratios may therefore provide conflicting evaluations of the same asset, especially across varying market regimes.
The objective of this project is to systematically examine these pros-cons and to assess whether a complementary approach based on exponential regression can address some of the limitations inherent in traditional risk-adjusted measures by focusing on the stability of the return-generation process rather than on a single risk proxy.

2.2. Related Work:

2.2.1. Improvements and Alternatives to Traditional Ratios

Many researchers have noted that classical ratios (Sharpe, Sortino, etc.) do not capture the full complexity of financial risk. In response, several alternative measures were proposed:

- **Omega Ratio** - considers the entire return distribution rather than a single volatility measure.
- **Upside Potential Ratio** - focuses on the frequency and magnitude of returns above a target level.

Key insight:
These developments highlight a consensus in the literature:
volatility alone is not a complete measure of risk, and additional structural information is needed.

This reinforces the motivation for our study: understanding where classical ratios succeed or fail.

2.2.2 Compounding Effects and Exponential Price Behavior

Financial asset prices evolve multiplicatively, not additively.
Therefore, many works model prices using:

- log-returns
- log-linear regressions

These models have shown that:

- Stable compounding often appears as a near-linear curve in log-scale.
- Instability, regime shifts, or inconsistent return patterns weaken the exponential fit.

Relevance to our project:
This provides theoretical support for our proposed metric, which evaluates how consistently an asset follows an exponential compounding structure, and not simply whether it trends upward.

2.2.3 Behavior of ratios under trending markets

Empirical evidence indicates that risk-adjusted performance measures exhibit regime-dependent behavior and respond differently across distinct market phases. Jegadeesh and Titman (1993) demonstrate that assets experiencing sustained price trends tend to display strong momentum effects, which can lead volatility-based ratios such as the Sharpe and Sortino ratios to overestimate performance during prolonged growth periods.

Moskowitz, Ooi, and Pedersen (2012) further show that time-series momentum strategies perform well during stable trending regimes but suffer sharp reversals during abrupt market transitions, highlighting the sensitivity of return-based metrics to structural changes in trend direction.

Additionally, Fama and French (2012) document that momentum-related effects vary significantly across market conditions, reinforcing the notion that performance ratios implicitly react to trend structure even when trend information is not explicitly modeled.

These findings support the need to distinguish between explicit momentum indicators and the implicit sensitivity of classical risk-adjusted ratios to underlying trend stability, thereby motivating the exploration of alternative measures that directly account for compounding structure rather than volatility alone.

2.2.4 Regression-Based Analysis of Price Structure

Several studies employ regression-based techniques to analyze the structural properties of financial price series, focusing on smoothness, noise levels, and coherence of growth patterns. Linear and log-linear regression models are commonly used to estimate average growth rates and assess the stability of return-generating processes over time (Campbell, Lo, & MacKinlay, 1997).

Log-linear regressions applied to price levels allow researchers to model exponential growth behavior, which naturally arises from compounding returns. A strong regression fit in log-price space indicates stable compounding dynamics, whereas a weak fit often signals increased noise, regime shifts, or structural breaks in the underlying price process (Tsay, 2010).

Regression diagnostics, including goodness-of-fit measures and residual analysis, have also been used to detect deviations from expected growth patterns and to identify periods of elevated uncertainty or instability (Brockwell & Davis, 2016).

Relevance to our project:
Our proposed exponential-fit metric follows this line of research by using log-linear regression not as a forecasting tool, but as a mechanism for quantifying the consistency of the return-generation process itself. Rather than predicting future prices, the regression fit serves as an indicator of structural stability in historical compounding behavior.

2.2.5 Limitations of Classical Ratios in Academic Literature

The academic literature documents multiple structural limitations of classical risk-adjusted performance ratios. One widely recognized issue is volatility clustering, whereby periods of high and low volatility tend to persist, violating the assumption of constant variance underlying ratios such as Sharpe and Sortino (Engle, 1982).

Additionally, empirical return distributions exhibit fat tails and excess kurtosis, leading to a higher frequency of extreme events than predicted by normal distribution assumptions. This causes volatility-based ratios to systematically underestimate downside risk (Cont, 2001).

Drawdown-based concerns have also been highlighted, as large market crashes disproportionately affect investor utility compared to smaller fluctuations. While measures such as the Calmar ratio partially address this issue, they still fail to capture intermediate structural instability between peak and trough events (Magdon-Ismail et al., 2004).

Furthermore, several studies show that risk-adjusted ratios behave inconsistently across market regimes, producing materially different rankings of assets during bull markets, bear markets, and crisis periods (Kritzman & Li, 2010; Daniel & Moskowitz, 2016).

These findings collectively justify the motivation of this project: to explore whether incorporating additional structural information—such as the stability of exponential compounding—can complement classical ratios and improve risk-adjusted evaluation under non-stationary market conditions.

3. Methodology:

The methodology described here represents the planned workflow for the next phase of the project. Actual empirical evaluation will be carried out later.

We plan to:

1. **Compute and analyze traditional risk–return ratios**, using our implemented scripts and Yahoo Finance data. The goal is to study their behavior, compare their outputs across assets, and identify cases where the ratios may provide misleading signals.
2. **Review theoretical motivations and limitations** of each ratio, connecting mathematical assumptions to empirical outcomes.
3. **Develop the proposed exponential-fit metric**, which evaluates the degree to which historical prices conform to an exponential growth model.
4. **Compare the stability and interpretability** of the exponential-fit measure with established ratios.
5. **Identify scenarios for future empirical testing**, including different market regimes, asset classes, and stress periods.

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